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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Circular Motion

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	68 questions
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MCQ (Single Correct Answer) 68 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A car moving with a speed of $54\ \mathrm{km/h}$ takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $g = 10\ \mathrm{m/s^2}$)

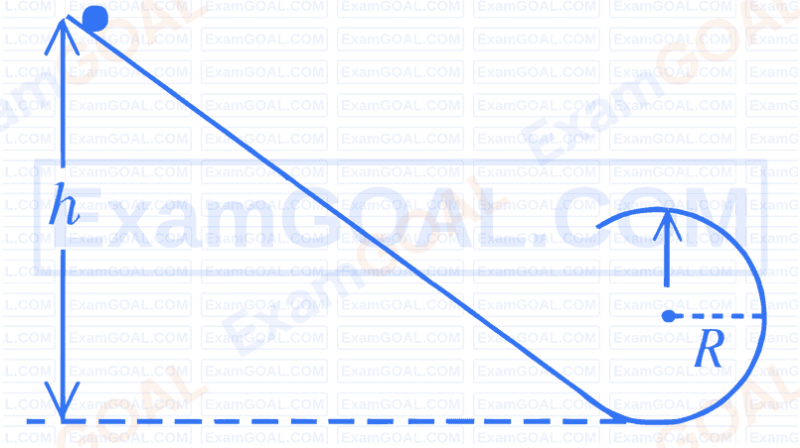
Open solution

(A) $\tan^{-1}(0.5)$ (B) $\tan^{-1}(0.75)$ (C) $\tan^{-1}(1.125)$ (D) $\tan^{-1}(0.25)$

Q2. JEE Main 2026 (Online) 6th April Morning Shift

A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height h as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height $h = \alpha R$. The value of α is ____

Open solution



(A) 2 (B) 4 (C) 3 (D) 6

Q3. JEE Main 2026 (Online) 2nd April Evening Shift

A 0.5 kg mass is in contact against the inner wall of a cylindrical drum of radius 4 m rotating about its vertical axis. The minimum rotational speed of the drum to enable the mass to remain stuck to the wall (without falling) is 5 rad/s. The coefficient of friction between the drum's inner wall surface and mass is _____. (Take $g = 10\ \mathrm{m/s^2}$)

Open solution

(A) 0.1 (B) 0.5 (C) 0.7 (D) 0.3

Q4. JEE Main 2026 (Online) 2nd April Morning Shift

A particle is rotating in a circular path and at any instant its motion can be described as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$.

The angular acceleration of the particle after 10 seconds is _____ rad/s^2.

Open solution

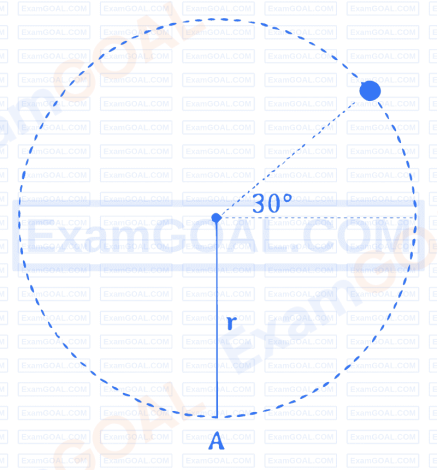
(A) 150 (B) 120 (C) 130 (D) 170

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Q5. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is (g = gravitational acceleration)

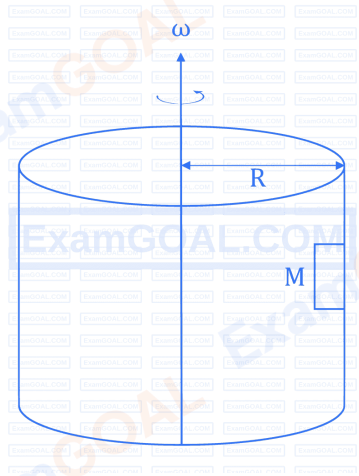


(A) $\sqrt{\frac{7}{2}gr}$ (B) $\sqrt{4gr}$ (C) $\sqrt{\frac{5}{2}gr}$ (D) $\sqrt{5gr}$

Q6. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

A large drum having radius R is spinning around its axis with angular velocity ω, as shown in figure. The minimum value of ω so that a body of mass M remains stuck to the inner wall of the drum, taking the coefficient of friction between the drum surface and mass M as μ, is :



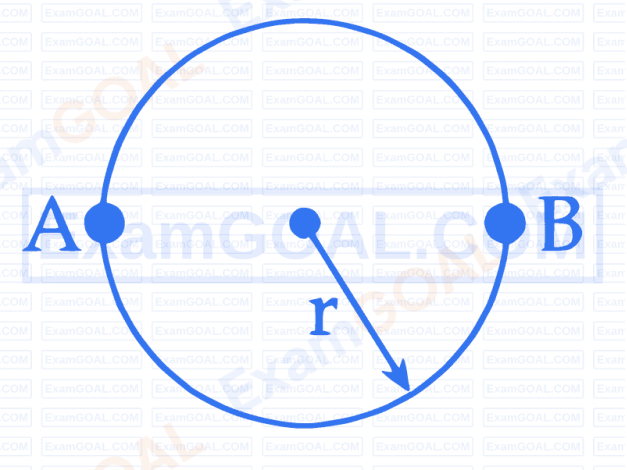
(A) $\sqrt{\frac{2g}{\mu R}}$ (B) $\sqrt{\frac{g}{2\mu R}}$ (C) $\sqrt{\frac{\mu g}{R}}$ (D) $\sqrt{\frac{g}{\mu R}}$

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Q7. JEE Main 2025 (Online) 2nd April Evening Shift

A sportsman runs around a circular track of radius r such that he traverses the path $ABAB$. The distance travelled and displacement, respectively, are

Open solution



(A) $\pi r, 3r$ (B) $2\mathrm{r}, 3\pi\mathrm{r}$ (C) $3\pi\mathrm{r}, 2\mathrm{r}$ (D) $3\pi r, \pi r$

Q8. JEE Main 2025 (Online) 29th January Morning Shift

A body of mass 'm' connected to a massless and unstretchable string goes in vertical circle of radius 'R' under gravity g. The other end of the string is fixed at the center of circle. If velocity at top of circular path is $n\sqrt{gR}$, where, n ≥ 1, then ratio of kinetic energy of the body at bottom to that at top of the circle is :

Open solution

(A) $\frac{n^2+4}{n^2}$ (B) $\frac{n+4}{n}$ (C) $\frac{n^2}{n^2+4}$ (D) $\frac{n}{n+4}$

Q9. JEE Main 2025 (Online) 24th January Morning Shift

A car of mass ' m ' moves on a banked road having radius ' r ' and banking angle θ. To avoid slipping from banked road, the maximum permissible speed of the car is v_0. The coefficient of friction μ between the wheels of the car and the banked road is

Open solution

(A) $\mu = \frac{v_0^2+rg\tan\theta}{rg+v_0^2\tan\theta}$

(B) $\mu = \frac{v_0^2-rg\tan\theta}{\mathrm{rg}-\mathrm{v_o^2}\tan\theta}$

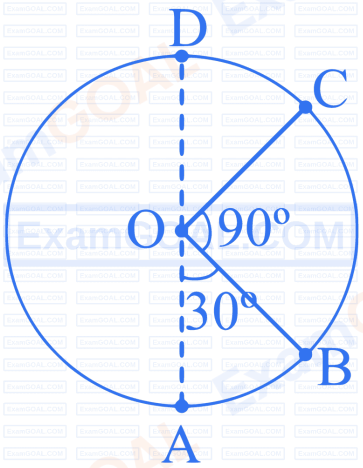
(C) $\mu = \frac{v_0^2-rg\tan\theta}{rg+v_0^2\tan\theta}$

(D) $\mu = \frac{v_o^2+rg\tan\theta}{rg-v_o^2\tan\theta}$

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Q10. JEE Main 2025 (Online) 22nd January Evening Shift

A body of mass 100 g is moving in circular path of radius 2 m on vertical plane as shown in figure. The velocity of the body at point A is $10\ \mathrm{m/s}$. The ratio of its kinetic energies at point B and C is :



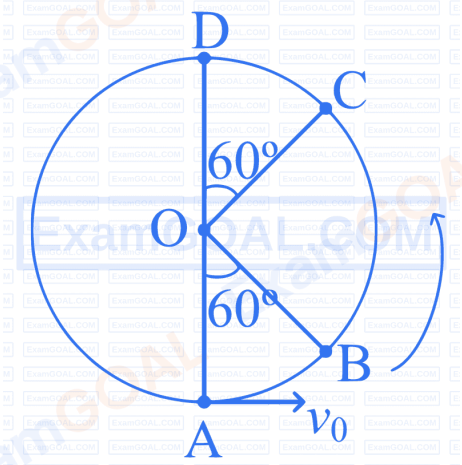
(Take acceleration due to gravity as $10\ \mathrm{m/s^2}$)

(A) $\frac{3-\sqrt{2}}{2}$ (B) $\frac{2+\sqrt{3}}{3}$ (C) $\frac{2+\sqrt{2}}{3}$ (D) $\frac{3+\sqrt{3}}{2}$

Open solution

Q11. JEE Main 2025 (Online) 22nd January Morning Shift

A bob of mass m is suspended at a point O by a light string of length l and left to perform vertical motion (circular) as shown in figure. Initially, by applying horizontal velocity v_0 at the point ' A ', the string becomes slack when, the bob reaches at the point ' D '. The ratio of the kinetic energy of the bob at the points B and C is _____.



(A) 4 (B) 1 (C) 2 (D) 3

Open solution

Q12. JEE Main 2024 (Online) 8th April Morning Shift

A clock has $75\ \mathrm{cm}, 60\ \mathrm{cm}$ long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take $\pi = 3.14$) :

(A) 118.9 (B) 140.5 (C) 139.4 (D) 220.0

Open solution

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Q13. JEE Main 2024 (Online) 6th April Evening Shift

A car of $800\ \mathrm{kg}$ is taking turn on a banked road of radius $300\ \mathrm{m}$ and angle of banking 30°. If coefficient of static friction is 0.2 then the maximum speed with which car can negotiate the turn safely: $(\mathrm{g} = 10\ \mathrm{m/s^2}, \sqrt{3} = 1.73)$

Open solution

(A) 51.4 m/s (B) 102.8 m/s (C) 70.4 m/s (D) 264 m/s

Q14. JEE Main 2024 (Online) 5th April Evening Shift

A man carrying a monkey on his shoulder does cycling smoothly on a circular track of radius $9\ \mathrm{m}$ and completes 120 revolutions in 3 minutes. The magnitude of centripetal acceleration of monkey is (in $\mathrm{m/s^2}$) :

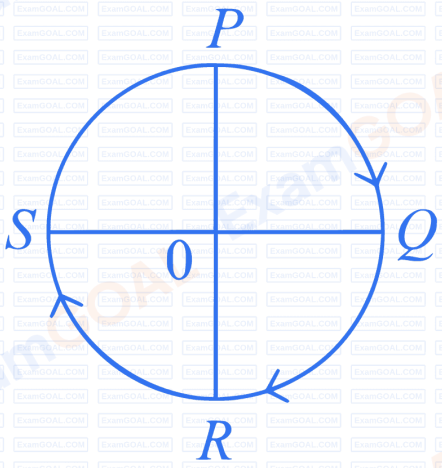
Open solution

(A) $4\pi^2\ \mathrm{ms}^{-2}$ (B) $16\pi^2\ \mathrm{ms}^{-2}$ (C) $57600\pi^2\ \mathrm{ms}^{-2}$ (D) Zero

Q15. JEE Main 2024 (Online) 4th April Evening Shift

A cyclist starts from the point P of a circular ground of radius $2\ \mathrm{km}$ and travels along its circumference to the point S. The displacement of a cyclist is:

Open solution



(A) $\sqrt{8}$ km (B) 4 km (C) 6 km (D) 8 km

Q16. JEE Main 2024 (Online) 1st February Morning Shift

A ball of mass $0.5\ \mathrm{kg}$ is attached to a string of length $50\ \mathrm{cm}$. The ball is rotated on a horizontal circular path about its vertical axis. The maximum tension that the string can bear is $400\ \mathrm{N}$. The maximum possible value of angular velocity of the ball in $\mathrm{rad/s}$ is :

Open solution

(A) 1600 (B) 20 (C) 40 (D) 1000

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Q17. JEE Main 2024 (Online) 1st February Morning Shift

A particle moving in a circle of radius R with uniform speed takes time T to complete one revolution.

If this particle is projected with the same speed at an angle θ to the horizontal, the maximum height attained by it is equal to $4R$. The angle of projection θ is then given by :

(A) $\sin^{-1}\left[\frac{2\mathrm{gT}^2}{\pi^2\mathrm{R}}\right]^{\frac{1}{2}}$

(B) $\sin^{-1}\left[\frac{\pi^2\mathrm{R}}{2\mathrm{gT}^2}\right]^{\frac{1}{2}}$

(C) $\cos^{-1}\left[\frac{\pi\mathrm{R}}{2\mathrm{gT}^2}\right]^{\frac{1}{2}}$

(D) $\cos^{-1}\left[\frac{2\mathrm{gT}^2}{\pi^2\mathrm{R}}\right]^{\frac{1}{2}}$

Open solution

Q18. JEE Main 2024 (Online) 31st January Morning Shift

A coin is placed on a disc. The coefficient of friction between the coin and the disc is μ. If the distance of the coin from the center of the disc is r, the maximum angular velocity which can be given to the disc, so that the coin does not slip away, is :

(A) $\sqrt{\frac{r}{\mu g}}$

(B) $\sqrt{\frac{\mu g}{r}}$

(C) $\frac{\mu g}{r}$

(D) $\frac{\mu}{\sqrt{rg}}$

Open solution

Q19. JEE Main 2024 (Online) 29th January Evening Shift

A stone of mass $900\ \mathrm{g}$ is tied to a string and moved in a vertical circle of radius $1\ \mathrm{m}$ making $10\ \mathrm{rpm}$. The tension in the string, when the stone is at the lowest point is (if $\pi^2 = 9.8$ and $g = 9.8\ \mathrm{m/s^2}$) :

(A) 17.8 N

(B) 97 N

(C) 9.8 N

(D) 8.82 N

Open solution

Q20. JEE Main 2024 (Online) 29th January Morning Shift

If the radius of curvature of the path of two particles of same mass are in the ratio $3 : 4$, then in order to have constant centripetal force, their velocities will be in the ratio of :

(A) $1 : \sqrt{3}$

(B) $2 : \sqrt{3}$

(C) $\sqrt{3} : 2$

(D) $\sqrt{3} : 1$

Open solution

Q21. JEE Main 2024 (Online) 27th January Morning Shift

A train is moving with a speed of $12\ \mathrm{m/s}$ on rails which are $1.5\ \mathrm{m}$ apart. To negotiate a curve of radius $400\ \mathrm{m}$, the height by which the outer rail should be raised with respect to the inner rail is (Given, $g = 10\ \mathrm{m/s^2}$) :

(A) 6.0 cm

(B) 5.4 cm

(C) 4.8 cm

(D) 4.2 cm

Open solution

Q22. JEE Main 2023 (Online) 13th April Evening Shift

A vehicle of mass $200\ \mathrm{kg}$ is moving along a levelled curved road of radius $70\ \mathrm{m}$ with angular velocity of $0.2\ \mathrm{rad/s}$. The centripetal force acting on the vehicle is:

(A) $560\ \mathrm{N}$

(B) $14\ \mathrm{N}$

(C) $2800\ \mathrm{N}$

(D) $2240\ \mathrm{N}$

Open solution

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Q23. JEE Main 2023 (Online) 11th April Morning Shift

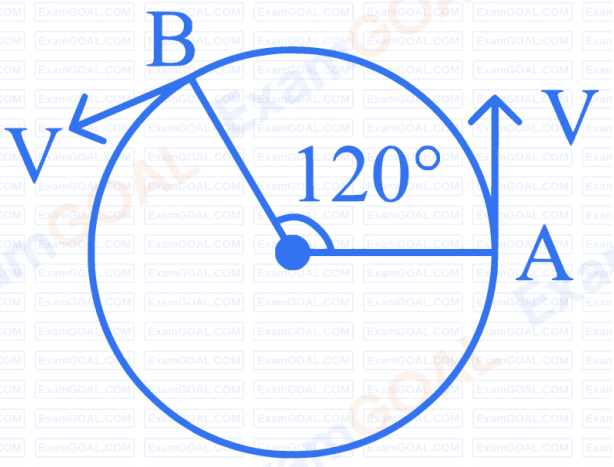
A coin placed on a rotating table just slips when it is placed at a distance of $1\ \text{cm}$ from the center. If the angular velocity of the table is halved, it will just slip when placed at a distance of _____ from the centre :

Open solution

(A) 1 cm (B) 8 cm (C) 4 cm (D) 2 cm

Q24. JEE Main 2023 (Online) 6th April Evening Shift

As shown in the figure, a particle is moving with constant speed $\pi\ \text{m/s}$. Considering its motion from A to B, the magnitude of the average velocity is :



Open solution

(A) $\pi\ \text{m/s}$ (B) $1.5\sqrt{3}\ \text{m/s}$
(C) $\sqrt{3}\ \text{m/s}$ (D) $2\sqrt{3}\ \text{m/s}$

Q25. JEE Main 2023 (Online) 6th April Evening Shift

A child of mass $5\ \text{kg}$ is going round a merry-go-round that makes 1 rotation in $3.14\ \text{s}$. The radius of the merry-go-round is $2\ \text{m}$. The centrifugal force on the child will be

Open solution

(A) 50 N (B) 80 N (C) 100 N (D) 40 N

Q26. JEE Main 2023 (Online) 6th April Morning Shift

A particle is moving with constant speed in a circular path. When the particle turns by an angle 90°, the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be -

Open solution

(A) 1 (B) 7 (C) 5 (D) 2

Q27. JEE Main 2023 (Online) 6th April Morning Shift

A small block of mass $100\ \text{g}$ is tied to a spring of spring constant $7.5\ \text{N/m}$ and length $20\ \text{cm}$. The other end of spring is fixed at a particular point A. If the block moves in a circular path on a smooth horizontal surface with constant angular velocity $5\ \text{rad/s}$ about point A, then tension in the spring is -

Open solution

(A) 0.50 N (B) 1.5 N (C) 0.75 N (D) 0.25 N

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Q28. JEE Main 2023 (Online) 31st January Evening Shift

A stone of mass $1\ \text{kg}$ is tied to end of a massless string of length $1\ \text{m}$. If the breaking tension of the string is $400\ \text{N}$, then maximum linear velocity, the stone can have without breaking the string, while rotating in horizontal plane, is :

Open solution

(A) $20\ \text{ms}^{-1}$ (B) $40\ \text{ms}^{-1}$ (C) $400\ \text{ms}^{-1}$ (D) $10\ \text{ms}^{-1}$

Q29. JEE Main 2023 (Online) 31st January Evening Shift

A body is moving with constant speed, in a circle of radius $10\ \text{m}$. The body completes one revolution in $4\ \text{s}$. At the end of 3rd second, the displacement of body (in m) from its starting point is :

Open solution

(A) 15π (B) 30 (C) $10\sqrt{2}$ (D) 5π

Q30. JEE Main 2023 (Online) 29th January Evening Shift

An object moves at a constant speed along a circular path in a horizontal plane with center at the origin. When the object is at $x = +2\ \text{m}$, its velocity is $-4\hat{\text{j}}$ m/s. The object's velocity (v) and acceleration (a) at $x = -2\ \text{m}$ will be

Open solution

(A) $v = 4\hat{\text{i}}\ \text{m/s}, a = 8\hat{\text{j}}\ \text{m/s}^2$ (B) $v = 4\hat{\text{j}}\ \text{m/s}, a = 8\hat{\text{i}}\ \text{m/s}^2$

(C) $v = -4\hat{\text{i}}\ \text{m/s}, a = -8\hat{\text{j}}\ \text{m/s}^2$ (D) $v = -4\hat{\text{j}}\ \text{m/s}, a = 8\hat{\text{i}}\ \text{m/s}^2$

Q31. JEE Main 2023 (Online) 29th January Morning Shift

A car is moving on a horizontal curved road with radius 50 m. The approximate maximum speed of car will be, if the coefficient of friction between tyres and road is 0.34. [take g = 10 ms^{-2}]

Open solution

(A) 3.4 ms^{-1} (B) 13 ms^{-1} (C) 22.4 ms^{-1} (D) 17 ms^{-1}

Q32. JEE Main 2023 (Online) 25th January Morning Shift

A car is moving with a constant speed of 20 m/s in a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car by a massless string. The angle made by the string with the vertical will be : (Take g = 10 m/s^2)

Open solution

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Q33. JEE Main 2023 (Online) 24th January Evening Shift

A body of mass 200g is tied to a spring of spring constant 12.5 N/m, while the other end of spring is fixed at point O. If the body moves about O in a circular path on a smooth horizontal surface with constant angular speed 5 rad/s, then the ratio of extension in the spring to its natural length will be :

Open solution

(A) 1 : 2 (B) 2 : 3 (C) 2 : 5 (D) 1 : 1

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Q34. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

A mass of $0.5\ \mathrm{kg}$ is attached to one end of a massless spring whose natural length is $1\ \mathrm{m}$ and spring constant is $800\ \mathrm{N/m}$ The other end of the spring is fixed while the mass moves in a circular path in a horizontal plane with an angular speed of $20\ \mathrm{rad/s}$. The extension in the length of the spring will be :

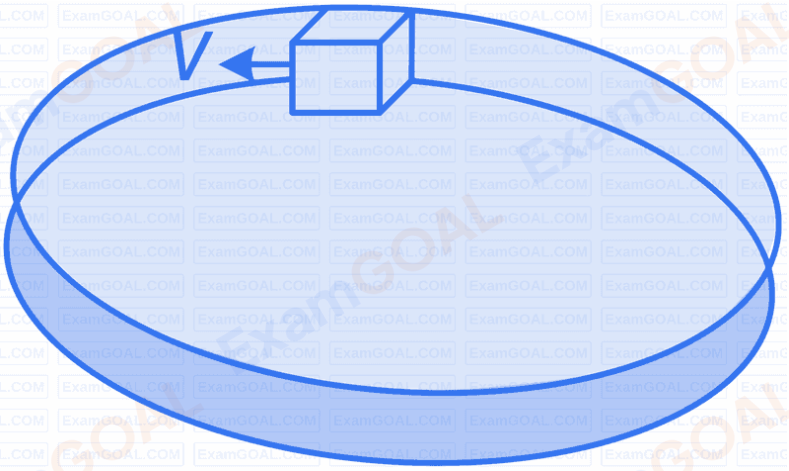
(A) $\frac{1}{10}\ \mathrm{m}$ (B) $\frac{1}{4}\ \mathrm{m}$ (C) $\frac{1}{3}\ \mathrm{m}$ (D) $\frac{1}{5}\ \mathrm{m}$

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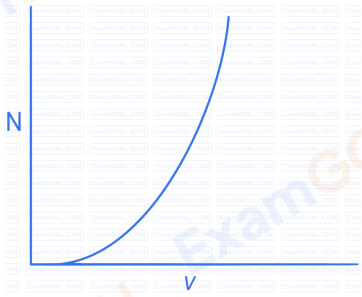
Q35. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

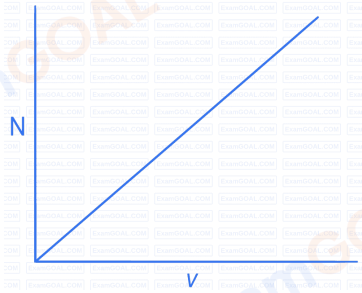
A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



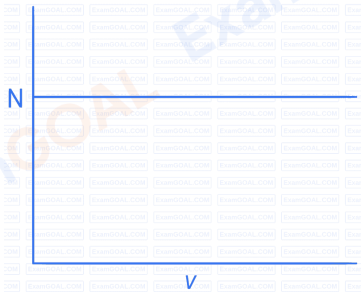
(A)



(B)

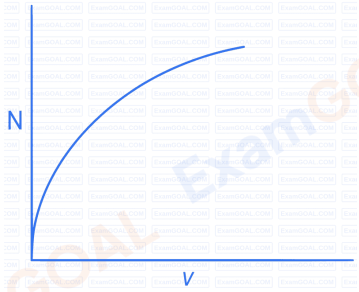


(C)



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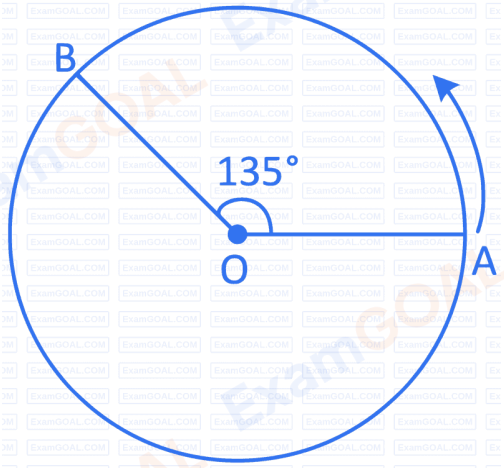
(D)



Q36. JEE Main 2022 (Online) 25th July Morning Shift

A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is $60\ \mathrm{m}$, then the magnitude of displacement would be :

(Given $\cos 135^\circ = -0.7$)



Open solution

(A) 42 m (B) 47 m (C) 19 m (D) 40 m

Q37. JEE Main 2022 (Online) 28th June Morning Shift

A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration (a) is varying with time t as $a = k^2rt^2$, where k is a constant. The power delivered to the particle by the force acting on it is given as

Open solution

(A) zero (B) $mk^2r^2t^2$ (C) mk^2r^2t (D) mk^2rt

Q38. JEE Main 2022 (Online) 27th June Evening Shift

A stone tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2 - gL)}$. The value of x is -

Open solution

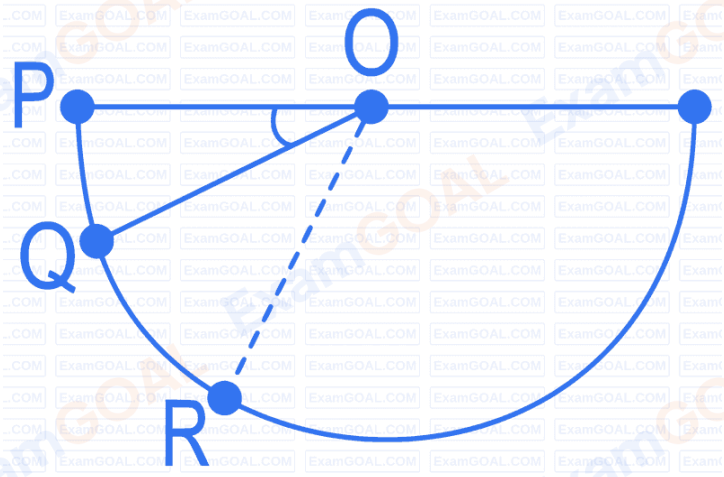
(A) 3 (B) 2 (C) 1 (D) 5

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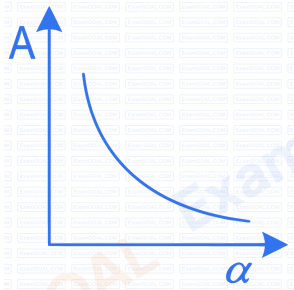
Q39. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

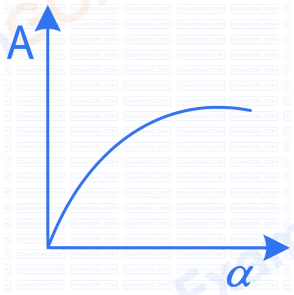
A ball is released from rest from point P of a smooth semi-spherical vessel as shown in figure. The ratio of the centripetal force and normal reaction on the ball at point Q is A while angular position of point Q is α with respect to point P. Which of the following graphs represent the correct relation between A and α when ball goes from Q to R?



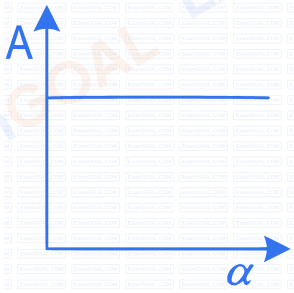
(A)



(B)

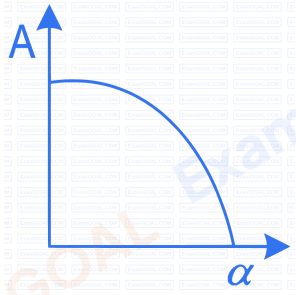


(C)



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(D)



Q40. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if :

(A) $R \leq \frac{\mu g}{2\omega^2}$

(B) $R \leq \frac{\mu g}{\omega^2}$

(C) $R \geq \frac{\mu g}{2\omega^2}$

(D) $R \geq \frac{\mu g}{\omega^2}$

Q41. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

For a particle in uniform circular motion, the acceleration $\vec{a}$ at any point P(R, θ) on the circular path of radius R is (when θ is measured from the positive x-axis and v is uniform speed) :

(A) $-\frac{v^2}{R}\sin\theta\hat{i} + \frac{v^2}{R}\cos\theta\hat{j}$

(B) $-\frac{v^2}{R}\cos\theta\hat{i} + \frac{v^2}{R}\sin\theta\hat{j}$

(C) $-\frac{v^2}{R}\cos\theta\hat{i} - \frac{v^2}{R}\sin\theta\hat{j}$

(D) $-\frac{v^2}{R}\hat{i} + \frac{v^2}{R}\hat{j}$

Q42. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

A stone of mass m, tied to a string is being whirled in a vertical circle with a uniform speed. The tension in the string is

(A) the same throughout the motion.

(B) minimum at the highest position of the circular path.

(C) minimum at the lowest position of the circular path.

(D) minimum when the rope is in the horizontal position.

Q43. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

A boy ties a stone of mass 100 g to the end of a 2 m long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of 80 N. If the maximum speed with which the stone can revolve is $\frac{K}{\pi}$ rev./min. The value of K is :

(Assume the string is massless and unstretchable)

(A) 400 (B) 300 (C) 600 (D) 800

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Q44. JEE Main 2021 (Online) 27th August Morning Shift

A huge circular arc of length 4.4 ly subtends an angle of 4" (4 seconds of arc) at the centre of the circle. How long it would take for a body to complete 4 revolutions if its speed is 8 AU per second?

Open solution

Given : 1 ly = 9.46×10^{15} m

1 AU = 1.5×10^{11} m

(A) 4.1×10^{8} s (B) 4.5×10^{10} s (C) 3.5×10^{6} s (D) 7.2×10^{8} s

Q45. JEE Main 2021 (Online) 26th August Evening Shift

A particle of mass m is suspended from a ceiling through a string of length L. The particle moves in a horizontal circle of radius r such that $r = \frac{L}{\sqrt{2}}$. The speed of particle will be :

Open solution

(A) $\sqrt{rg}$ (B) $\sqrt{2rg}$ (C) $2\sqrt{rg}$ (D) $\sqrt{\frac{rg}{2}}$

Q46. JEE Main 2021 (Online) 20th July Morning Shift

The normal reaction 'N' for a vehicle of 800 kg mass, negotiating a turn on a 30° banked road at maximum possible speed without skidding is _____ $\times 10^3$ kg m/s^2. [Given cos30° = 0.87, μ_s = 0.2]

Open solution

(A) 12.4 (B) 7.2 (C) 6.96 (D) 10.2

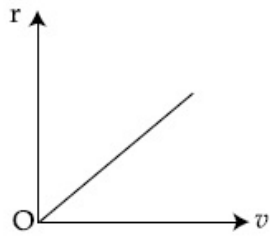
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Q47. JEE Main 2021 (Online) 18th March Evening Shift

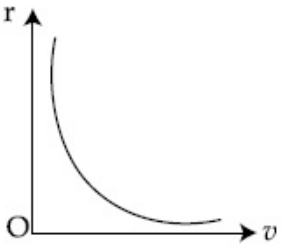
Open solution

A particle of mass m moves in a circular orbit under the central potential field, $U(r) = -\frac{C}{r}$, where C is a positive constant. The correct radius – velocity graph of the particle's motion is :

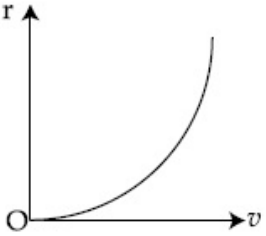
(A)



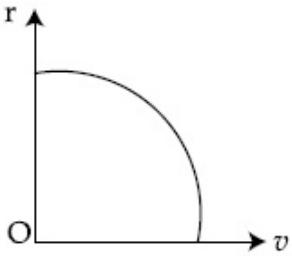
(B)



(C)



(D)

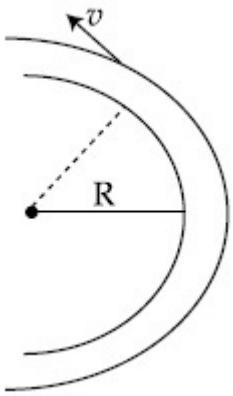


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Q48. JEE Main 2021 (Online) 17th March Morning Shift

Open solution

A modern grand - prix racing car of mass m is travelling on a flat track in a circular arc of radius R with a speed v. If the coefficient of static friction between the tyres and the track is μ_s, then the magnitude of negative lift F_L acting downwards on the car is : (Assume forces on the four tyres are identical and g = acceleration due to gravity)



(A) $m\left(g-\frac{v^2}{\mu_s R}\right)$

(B) $-m\left(g+\frac{v^2}{\mu_s R}\right)$

(C) $m\left(\frac{v^2}{\mu_s R}-g\right)$

(D) $m\left(\frac{v^2}{\mu_s R}+g\right)$

Q49. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

Statement I : A cyclist is moving on an unbanked road with a speed of 7 kmh^{-1} and takes a sharp circular turn along a path of radius of 2m without reducing the speed. The static friction coefficient is 0.2. The cyclist will not slip and pass the curve. (g = 9.8 m/s^2)

Statement II : If the road is banked at an angle of 45°, cyclist can cross the curve of 2m radius with the speed of 18.5 kmh^{-1} without slipping.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is incorrect and statement II is correct

(B) Both statement I and statement II are true

(C) Statement I is correct and statement II is incorrect

(D) Both statement I and statement II are false

Q50. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

A block of 200 g mass moves with a uniform speed in a horizontal circular groove, with vertical side walls of radius 20 cm. If the block takes 40 s to complete one round, the normal force by the side walls of the groove is :

(A) 9.859×10^{-2} N

(B) 0.0314 N

(C) 9.859×10^{-4} N

(D) 6.28×10^{-3} N

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Q51. JEE Main 2021 (Online) 26th February Morning Shift

A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3. Its time period of revolution will be given by :

Open solution

(A) $T \propto R^{\frac{4}{3}}$ (B) $T \propto R^{\frac{5}{2}}$ (C) $T \propto R^{\frac{3}{2}}$ (D) $T \propto R^2$

Q52. JEE Main 2020 (Online) 6th September Morning Slot

A clock has a continuously moving second's hand of 0.1 m length. The average acceleration of the tip of the hand (in units of ms^{-2}) is of the order of :

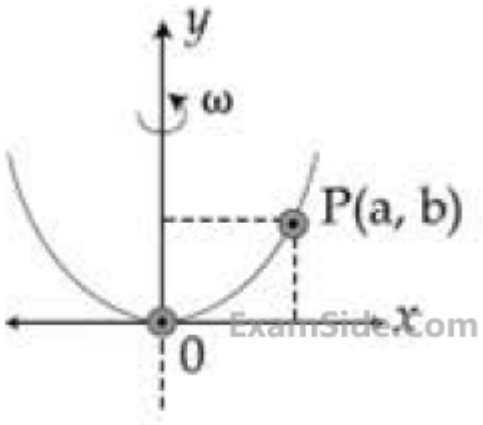
Open solution

(A) 10^{-3} (B) 10^{-1} (C) 10^{-2} (D) 10^{-4}

Q53. JEE Main 2020 (Online) 2nd September Morning Slot

A bead of mass m stays at point P(a, b) on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure). The value of ω is (neglect friction) :

Open solution



(A) $2\sqrt{2gC}$ (B) $2\sqrt{gC}$ (C) $\sqrt{\frac{2gC}{ab}}$ (D) $\sqrt{\frac{2g}{C}}$

Q54. JEE Main 2020 (Online) 8th January Morning Slot

A particle of mass m is fixed to one end of a light spring having force constant k and unstretched length ℓ. The other end is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is :

Open solution

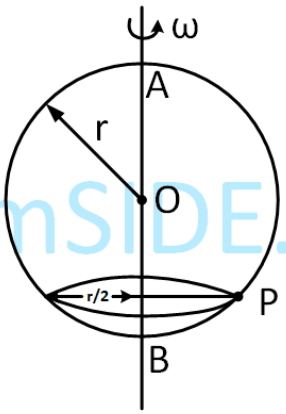
(A) $\frac{m\ell\omega^2}{k-m\omega^2}$ (B) $\frac{m\ell\omega^2}{k-m\omega}$
(C) $\frac{m\ell\omega^2}{k+m\omega^2}$ (D) $\frac{m\ell\omega^2}{k+m\omega}$

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Q55. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to -



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(A) $\frac{\sqrt{3}g}{2r}$

(B) $\frac{2g}{\left(r\sqrt{3}\right)}$

(C) $\frac{\left(g\sqrt{3}\right)}{r}$

(D) $\frac{2g}{r}$

Q56. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

A particle is moving along a circular path with a constant speed of 10 ms^{-1}. What is the magnitude of the change in velocity of the particle, when it moves through an angle of 60° around the centre of the circle?

(A) zero

(B) 10 m/s

(C) $10\sqrt{2}m/s$

(D) $10\sqrt{3}m/s$

Q57. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

A body is projected at t = 0 with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at t = 1s is R. neglecting air resistance and taking acceleration due to gravity g = 10 ms^{-2}, the value of R is :

(A) 2.8 m

(B) 5.1 m

(C) 2.5 m

(D) 10.3 m

Q58. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

A disc rotates about its axis of symmetry in a horizontal plane at a steady rate of 3.5 revolutions per second. A coin placed at a distance of 1.25 cm from the axis of rotation remains at rest on the disc. The coefficient of friction between the coin and the disc is : (g = 10 m/s^2)

(A) 0.5

(B) 0.3

(C) 0.7

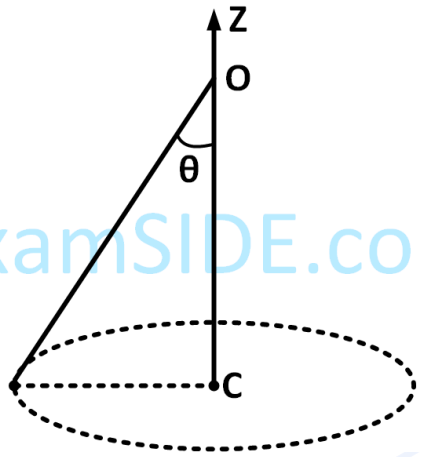
(D) 0.6

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Q59. JEE Main 2017 (Online) 9th April Morning Slot

A conical pendulum of length 1 m makes an angle $\theta = 45^{o}$ w.r.t. Z-axis and moves in a circle in the XY plane. The radius of the circle is 0.4 m and its center is vertically below O. The speed of the pendulum, in its circular path, will be: (Take g = 10 ms^{-2})

Open solution



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(A) 0.4 m/s (B) 4 m/s (C) 0.2 m/s (D) 2 m/s

Q60. JEE Main 2016 (Online) 10th April Morning Slot

Concrete mixture is made by mixing cement, stone and sand in a rotating cylindrical drum. If the drum rotates too fast, the ingredients remain stuck to the wall of the drum and proper mixing of ingredients does not take place. The maximum rotational speed of the drum in revolutions per minute(rpm) to ensure proper mixing is close to :

Open solution

(Take the radius of the drum to be 1.25 m and its axle to be horizontal) :

(A) 0.4 (B) 1.3 (C) 8.0 (D) 27.0

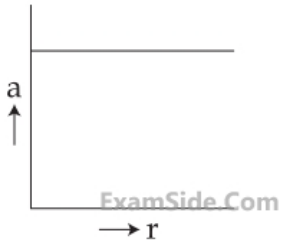
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Q61. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

If a body moving in a circular path maintains constant speed of $10ms^{-1}$, then which of the following correctly describes relation between acceleration and radius ?

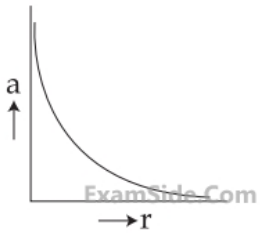
(A)



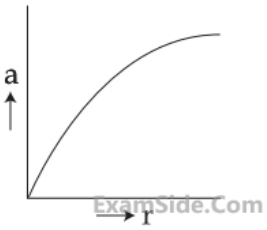
(B)



(C)

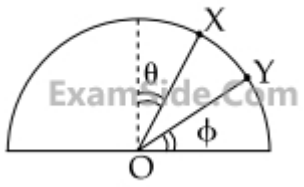


(D)



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Q62. JEE Main 2014 (Online) 19th April Morning Slot



Open solution

A particle is released on a vertical smooth semicircular track from point X so that OX makes angle θ from the vertical (see figure). The normal reaction of the track on the particle vanishes at point Y where OY makes angle ϕ with the horizontal. Then :

(A) $\sin\phi = \cos\theta$

(B) $\sin\phi = \frac{1}{2}\cos\theta$

(C) $\sin\phi = \frac{2}{3}\cos\theta$

(D) $\sin\phi = \frac{3}{4}\cos\theta$

Q63. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

A body of mass 'm' is tied to one end of a spring and whirled round in a horizontal plane with a constant angular velocity. The elongation in the spring is 1 cm. If the angular velocity is doubled, the elongation in the spring is 5 cm. The original length of the spring is :

(A) 15 cm

(B) 12 cm

(C) 16 cm

(D) 10 cm

Q64. AIEEE 2012

Open solution

Two cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2, respectively. Their speeds are such that they make complete circles in the same time t. The ratio of their centripetal acceleration is

(A) $m_1r_1 : m_2r_2$

(B) $m_1 : m_2$

(C) $r_1 : r_2$

(D) 1 : 1

Q65. AIEEE 2010

Open solution

For a particle in uniform circular motion the acceleration $\vec{a}$ at a point P(R, θ) on the circle of radius R is (here θ is measured from the x–axis)

(A) $-\frac{v^2}{R}\cos\theta\hat{i} + \frac{v^2}{R}\sin\theta\hat{j}$

(B) $-\frac{v^2}{R}\sin\theta\hat{i} + \frac{v^2}{R}\cos\theta\hat{j}$

(C) $-\frac{v^2}{R}\cos\theta\hat{i} - \frac{v^2}{R}\sin\theta\hat{j}$

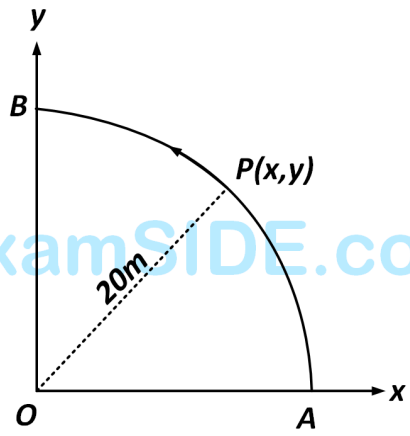
(D) $\frac{v^2}{R}\hat{i} + \frac{v^2}{R}\hat{j}$

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Q66. AIEEE 2010

A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of P is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and t is in seconds. The radius of the path is $20\,m$. The acceleration of $'P'$ when $t = 2\,s$ is nearly.

Open solution



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(A) $13m/s_2$ (B) $12m/s^2$ (C) $7.2ms^2$ (D) $14m/s^2$

Q67. AIEEE 2004

Which of the following statements is **FALSE** for a particle moving in a circle with a constant angular speed?

Open solution

(A) The velocity vector is tangent to the circle.

(B) The acceleration vector is tangent to the circle.

(C) The acceleration vector points to the centre of the circle.

(D) The velocity and acceleration vectors are perpendicular to each other.

Q68. AIEEE 2002

The maximum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is

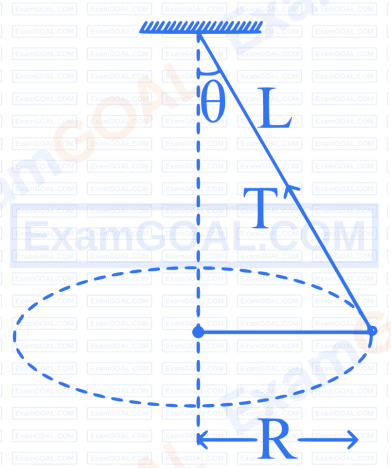
Open solution

(A) 60 (B) 30 (C) 15 (D) 25

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Numerical 9 questions

Q69. JEE Main 2025 (Online) 24th January Evening Shift



Open solution

A string of length L is fixed at one end and carries a mass of M at the other end. The mass makes $\left(\frac{3}{\pi}\right)$ rotations per second about the vertical axis passing through end of the string as shown. The tension in the string is _____ ML.

Q70. JEE Main 2025 (Online) 22nd January Evening Shift

A tube of length 1 m is filled completely with an ideal liquid of mass 2 M , and closed at both ends. The tube is rotated uniformly in horizontal plane about one of its ends. If the force exerted by the liquid at the other end is F then angular velocity of the tube is $\sqrt{\frac{\mathrm{F}}{\alpha \mathrm{M}}}$ in SI unit. The value of α is _____.

Open solution

Q71. JEE Main 2024 (Online) 29th January Evening Shift

A particle is moving in a circle of radius $50\ \mathrm{cm}$ in such a way that at any instant the normal and tangential components of it's acceleration are equal. If its speed at $\mathrm{t} = 0$ is $4\ \mathrm{m/s}$, the time taken to complete the first revolution will be $\frac{1}{\alpha}\left[1 - e^{-2\pi}\right]\mathrm{s}$, where $\alpha =$ _____.

Open solution

Q72. JEE Main 2023 (Online) 30th January Evening Shift

A stone tied to $180\ \mathrm{cm}$ long string at its end is making 28 revolutions in horizontal circle in every minute. The magnitude of acceleration of stone is $\frac{1936}{x} ms^{-2}$. The value of x _____. (Take $\pi = \frac{22}{7}$)

Open solution

Q73. JEE Main 2023 (Online) 29th January Evening Shift

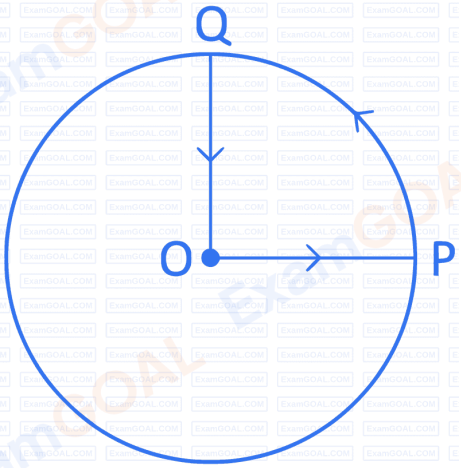
A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential acceleration and centripetal acceleration are equal. The time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1 - e^{-\pi/2})s$. The value of t is _____.

Open solution

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Q74. JEE Main 2022 (Online) 30th June Morning Shift

A person starts his journey from centre 'O' of the park and comes back to the same position following path OPQO as shown in the figure. The radius of path taken by the person is 200 m and he takes 3 min 58 sec to complete his journey. The average speed of the person is _____ ms^{-1}. (take π = 3.14)



Open solution

Q75. JEE Main 2022 (Online) 27th June Morning Shift

A pendulum of length 2 m consists of a wooden bob of mass 50 g. A bullet of mass 75 g is fired towards the stationary bob with a speed v. The bullet emerges out of the bob with a speed $\frac{v}{3}$ and the bob just completes the vertical circle. The value of v is _____ ms^{-1}. (if g = 10 m/s^2).

Open solution

Q76. JEE Main 2022 (Online) 25th June Evening Shift

A curve in a level road has a radius 75 m. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48 m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be _____ m/s.

Open solution

Q77. JEE Main 2021 (Online) 25th February Morning Shift

A small bob tied at one end of a thin string of length 1 m is describing a vertical circle so that the maximum and minimum tension in the string are in the ratio 5 : 1. The velocity of the bob at the highest position is _____ m/s. (Take g = 10 m/s^2)

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **A**	4. **C**	5. **A**
6. **D**	7. **C**	8. **A**	9. **C**	10. **D**
11. **C**	12. **C**	13. **A**	14. **B**	15. **A**
16. **C**	17. **A**	18. **B**	19. **C**	20. **C**
21. **B**	22. **A**	23. **C**	24. **B**	25. **D**
26. **D**	27. **C**	28. **A**	29. **C**	30. **B**
31. **B**	32. **C**	33. **B**	34. **C**	35. **A**
36. **B**	37. **C**	38. **B**	39. **C**	40. **B**
41. **C**	42. **B**	43. **C**	44. **B**	45. **A**
46. **D**	47. **B**	48. **C**	49. **B**	50. **C**
51. **D**	52. **A**	53. **A**	54. **A**	55. **B**
56. **B**	57. **A**	58. **D**	59. **D**	60. **D**
61. **C**	62. **C**	63. **A**	64. **C**	65. **C**
66. **D**	67. **B**	68. **B**		

Numerical

69. **36**	70. **1**	71. **8**	72. **125**	73. **40**
74. **3**	75. **10**	76. **24**	77. **5**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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